

Tier	Function / Component	Manufacturer	Manufacturer Part Number	Description / Specification	Quantity	Unit Cost	Total Cost	Interface
2	3-axis acceleration	Analog Devices	ADXL355BEZ-RL7	Low-noise 3-axis MEMS accelerometer	1	77.41	77.41	SPI / I2C; 14-CLCC
2	X/Y/Z angular rate sensing	Analog Devices	ADXRS453BRGZ	±300 deg/s digital single-axis MEMS gyroscope	3	90.57	271.71	SPI; SOIC
2	X/Y magnetic sensing	Honeywell Aerospace	HMC1002-TR	2-axis magnetoresistive sensor bridge	1	52.91	52.91	Analog bridge; 20-SOIC
2	Z magnetic sensing	Honeywell Aerospace	HMC1001-RC	1-axis magnetoresistive sensor bridge	1	70.88	70.88	Analog bridge
2	Magnetometer signal conditioning	Analog Devices	AD8421TRMZ-EP	Low-noise precision instrumentation amplifier	3	25.52	76.56	8-MSOP
2	High-resolution sensor ADC	Texas Instruments	ADS1258MRTCTEP	24-bit 16-channel delta-sigma ADC	1	59.91	59.91	SPI; 48-VQFN
2	ADC voltage reference	Texas Instruments	REF5025MDTEP	2.5-V precision voltage reference	1	18.95	18.95	8-SOIC
2	Barometric/static + differential pressure	Amphenol All Sensors	AUAV-L30D-M25125-RR-N	AUAV dual airspeed and altitude pressure sensor	1	51.37	51.37	SPI / I2C; 12-SMD
2	Outside air temperature	Honeywell	HEL-705-U-1-12-00	1 kΩ platinum RTD probe	1	47.72	47.72	Resistive analog
2	RTOS / sensor-processing MCU	Texas Instruments	S5LS20216ASPGEMEP	TMS570LS20216-EP Hercules MCU	1	55.49	55.49	144-LQFP; SPI; UART; JTAG
2	ARINC 429 interface	Holt Integrated Circuits	HI-35930PCTF	Dual ARINC 429 receiver / single transmitter IC	1	106	106	SPI; 44-QFN
3	Magnetometer set/reset drive	TBD	TBD	HMC1001/HMC1002 set-reset pulse driver network	1			Analog / power pulse
3	Magnetometer gain/filter network	TBD	TBD	Precision gain, bias, and anti-alias filtering components	1			Analog
3	RTD excitation / measurement	TBD	TBD	Precision RTD excitation and input conditioning circuit	1			Analog
3	MCU clock / reset	TBD	TBD	TMS570 oscillator, reset, and local watchdog support	1			Clock / reset
3	Local regulators	TBD	TBD	5 V / 3.3 V / MCU core power rails	1			Power
3	Pitot/static plumbing	TBD	TBD	Pneumatic fittings, tubing, and ports	1			Pneumatic
3	External connector	TBD	TBD	Rugged aircraft connector	1			Aircraft I/O